



IMSearch 2.0: Toward User-Centric and Efficient Interactive Multimedia Retrieval System

Duc-Tuan Luu^{1,2,3,4} , Khanh-An C. Quan^{1,2,3,4} , Duy-Ngoc Nguyen^{1,4},
Khanh-Linh Bui-Le^{1,4}, Nhat-Sang Doan^{1,4}, Minh-Duc Le-Ngo^{1,4},
Vinh-Tiep Nguyen^{1,4} , and Minh-Triet Tran^{2,3,4}

¹ University of Information Technology, VNU-HCM, Ho Chi Minh City, Vietnam
{tuanld, anqck, tiepvn}@uit.edu.vn,

{21522380, 21522284, 21522542, 21520195}@gm.uit.edu.vn

² University of Science, VNU-HCM, Ho Chi Minh City, Vietnam
tmtriet@fit.hcmus.edu.vn

³ John von Neumann Institute, VNU-HCM, Ho Chi Minh City, Vietnam

⁴ Vietnam National University, Ho Chi Minh City, Vietnam

Abstract. The rapid growth of the internet and technology has led to an exponential increase in the volume of information individuals must manage, which results in a rising demand for search engines to assess relevance quickly. Hence, information retrieval has garnered increasing attention within research communities. The Video Browser Showdown (VBS) is an annual competition that allows researchers to evaluate their methodologies against established benchmarks. In this work, we introduce IMSEARCH 2.0, an enhanced interactive video retrieval system for VBS 2025. This revised version features several improvements aimed at assisting users in effectively addressing multimedia retrieval tasks. Particularly, IMSEARCH 2.0 retains the core search function across multiple modalities, including text, automatic speech recognition (ASR) and optical character recognition (OCR), followed by a relevance feedback mechanism. Additionally, we develop a new semi-automatic keyframe selection, a colour-dominant search and replace the traditional CLIP embedding methods with a more effective model to achieve a more comprehensive visual representation. Finally, the user interface has been modified to be more user-centric and easy-to-use for all users.

Keywords: Video Browser Showdown · Interactive Video Retrieval System · Vision-Language Model

1 Introduction

Nowadays, with the continuous development of technology, a wide range of digital recording devices has become both accessible and affordable to the public. As a result, recording and uploading videos to multimedia platforms like Facebook or

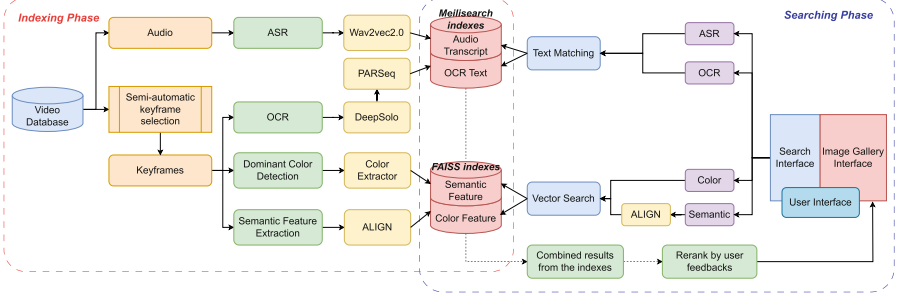


Fig. 1. The overall pipeline of our proposed IMSEARCH 2.0

TikTok has become remarkably easy. However, this ease of access has led to an explosion of visual and textual content, creating challenges for traditional search engines in managing and retrieving relevant information effectively. Thus, there is a growing demand for multimedia retrieval frameworks that can effectively balance the ability to process large-scale video data with the need for intuitive, user-friendly interfaces.

The annual Video Browser Showdown (VBS) [16] challenge is organized to promote a direct comparison of interactive video retrieval systems in a real-time setting using objective measurements. There are two main categories of video search tasks: Known-Item Search (KIS) and Adhoc-Video Search (AVS). KIS task requires users to search for a particular scene given an annotation-free multimedia collection (textual hints or short video segments). Meanwhile, for the AVS task, users need to retrieve as many frames as possible that are semantically relevant to the query descriptions. In 2024, VBS introduced a new challenge: Question and Answer (Q/A), where participants must identify the correct answers to questions from the extensive video corpus, further expanding the scope of multimedia retrieval.

In this paper, we present our multimedia retrieval framework, IMSEARCH 2.0, which is participating in VBS 2025. This interactive, large-scale video search system leverages off-the-shelf vision-language models (VLMs) and AI techniques for visual content analysis and advanced indexing methods to ensure efficient scalability. An overview of our system architecture is illustrated in Fig. 1.

2 System Description

Our IMSEARCH 2.0 system consists of several modules, each offering unique interactions tailored to specific purposes. Users have the flexibility to combine multiple modules to achieve more effective and contextual retrieval. For clarity and ease of understanding, we present each module individually in this section.

2.1 Semi-automatic Keyframe Selection

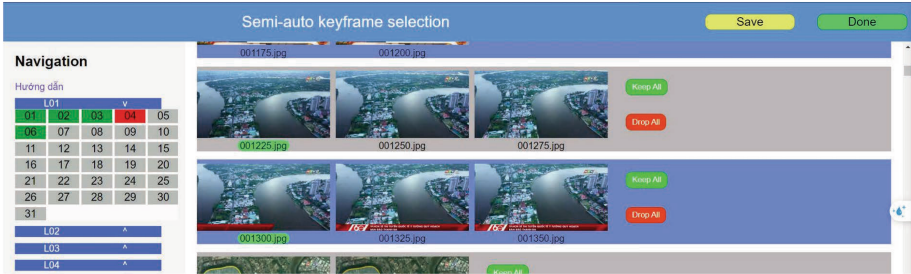


Fig. 2. Overview of the semi-automatic keyframe.

Traditionally, video frames are extracted based on frames-per-second (fps), often resulting in redundant images due to slow scene transitions. To address this, we first extract frames at 1fps and then deploy ALIGN [7] to generate feature embeddings. Cosine similarity measures frame similarity, with scores near 1.0 indicating similar content. Visually similar frames are grouped, and the most recent frame in each group is selected as the keyframe, while others are ignored. To minimize the risk of removing meaningful keyframes, we set a high threshold of 0.97 for cosine similarity. Despite this precaution, some redundant frames may still pass the threshold. To address this, system managers can review and manually retain keyframes as needed. Figure 2 illustrates this semi-automatic selection process. Users can also select individual images for retention, with unselected frames excluded from the database. This manual review is optional and can be applied selectively based on the user requirements.

2.2 Semantic Textual Search

In this module, we concentrate on efficiently addressing the fundamental requirements of video search. Being able to join the feature presentation for both visual and textual concepts, CLIP-based methods [10, 14] are widely employed across various research domains, such as image captioning or video understanding. Also, it has been used by several teams in the competition [1, 6, 11–13].

After conducting experiments with four models: CLIP [14], BLIP [10], ALIGN [7], and JINA [9] under the KIS setting using 35-query test sets from the Ho Chi Minh AI City Challenge dataset [5], we obtain the results shown in Table. 1. As observed, while the ALIGN model performs slightly lower in recall@1 compared to CLIP, it outperforms the other models across the remaining recall metrics. Given that recall@1 is not crucial in this competitive context, we use ALIGN for superior feature representation. The extracted features are then stored in FAISS database [8] which allows the use of cosine similarity for retrieval.

Table 1. Performance of image-text models with KIS setting.

	Recall@1	Recall@5	Recall@10	Recall@20
ALIGN [7]	0,29	0,49	0,63	0,69
BLIP [10]	0,26	0,31	0,37	0,43
JINA [9]	0,26	0,31	0,40	0,54
CLIP [14]	0,31	0,43	0,46	0,49

The **first and **second** best results are marked in **red** and **blue**.*

Additionally, to handle laparoscopic images from the challenging and domain-specific LapGynLHE dataset, relying on a general-purpose pre-trained model may be insufficient for effective processing. Therefore, we employ SurgVLP [20] pre-trained on hundreds of surgical video lectures to better adapt to the unique characteristics of the dataset.

2.3 ASR and OCR

Videos often contain audio speech, subtitles, scene texts, and document texts. If this data is properly processed, the functionality of our interactive multimedia search system may be enhanced. For speech recognition, our system utilizes Wave2vec 2.0 [3] to transcribe the audio in each video into textual subtitles. To extract text from images, we employ a standard OCR pipeline, specifically for scene text detection, which involves two phases. The first phase, text detection, uses the DeepSolo [19] model to locate text within the video frames. In the second phase, these identified regions undergo text recognition with the PARSeq [4] model to extract the textual content, producing text strings associated with the video frames. The extracted text from the OCR process and the transcribed subtitles from ASR are then indexed in two separate Meilisearch¹ databases, which support advanced text matching through fuzzy search algorithms.

2.4 Color-Dominant Search

Colour is widely regarded as a crucial aspect of human visual experience, playing a significant role in how we perceive and interpret images [17]. Leveraging this principle, we design a 5x5 grid layout that allows users to select a colour for each region, facilitating video content search based on their implicit memory of key visual events. To reduce the complexity of the RGB colour space and facilitate efficient retrieval, we employ Self-Organizing Maps (SOMs) [2] to quantize the colours. A single global SOM is trained using colour samples extracted from the entire image dataset, resulting in a set of prototype colour vectors that represent the quantized colour space. Each pixel in an image is then mapped to the closest prototype colour in the SOM, effectively reducing the continuous RGB space to a discrete set of representative colours. This quantization ensures consistent colour representation across all images. After quantization, we extract

¹ <https://www.meilisearch.com/>.

the dominant colours in each region within the 5x5 grid by identifying the most frequent quantized colours within each region. This captures the spatial distribution of colours, reflecting how people often recall visual details. We then deploy K-Means [18] clustering to group images that have similar dominant colours in corresponding regions. By combining SOM-based colour quantization with K-Means clustering, we efficiently organize the image database for rapid retrieval based on colour dominance. When a user inputs a query image, it is quantized using the pre-trained global SOM, and the dominant colours are extracted from each grid region. We then compare the colour descriptors of the query image with those of the clustered images in the database.

2.5 Re-rank by User Feedback

The initial search results often fall short of fully meeting user expectations, returning images that are either not entirely similar or dissimilar to the target content. To address this, we developed a re-ranking module based on the Ide Regular method [15], a foundational technique in relevance feedback. Particularly, the original search query, represented by vector q , is transformed into a modified query vector q_m by Eq. 1. This transformation uses user feedback to differentiate between relevant frames F_r and irrelevant frames F_i . Weights are assigned to the initial query (α), relevant video frames (β), and irrelevant video frames (γ), in which, $\alpha, \beta > 0$ and $\gamma < 0$. This iterative process continues until the most relevant result is achieved.

$$q_m = \alpha \cdot q + \beta \cdot \sum_{\forall f_i \in F_r} f_i + \gamma \cdot \sum_{\forall f_j \in F_i} f_j \quad (1)$$

2.6 User Interface

The IMSearch v2.0 interface forms the core of user interaction to provide a streamlined and smooth experience for users searching and retrieving relevant video frames (see Fig. 3). It features a two-column layout, where multiple search methods are integrated to offer flexibility and precision in the retrieval process. These search methods can be utilized individually or in combination, giving users the ability to fine-tune search results based on their specific needs. Once a query is submitted, a loading spinner is immediately displayed, giving users visual confirmation that the system is actively processing the search request.

Search Interface: This is the core interaction hub of our system, offering options like free text search, OCR/ASR search and colour patch filtering, which can be used individually or together for greater precision. A slider, located in the top-left corner, enables users to adjust the value of K, controlling the number of top-ranked frames displayed. Moreover, users can switch between different datasets, selecting the most relevant collection for their search needs.

Retrieval Interface: The retrieval interface presents the top K frames, ranked by confidence scores, accompanied by essential details such as frame name, a

button to enlarge the selected frame, a video player and a timeline preview. The timeline includes navigable arrows at the bottom, allowing users to browse adjacent frames efficiently. To maximize visibility while maintaining clarity, the interface employs a 5-column image grid that balances the number of displayed frames with adequate space for each frame’s details, ensuring the frames remain easily distinguishable and interactive. We also implement lazy loading to enhance performance, where only the images within the viewport are loaded initially, while additional frames are loaded dynamically as users scroll down. This optimization ensures faster loading times and smoother interactions. Furthermore, users can provide feedback on frame relevance through green (+) and red (-) buttons. This feedback mechanism directly contributes to improving search accuracy. Feedback frames are collected in the left sidebar, where users can review their marked frames and adjust rankings, enabling a more personalized search experience in future interactions.

The design of the IMSEARCH 2.0 system aims to be user-friendly and adaptable, providing smooth interaction while maintaining the flexibility to integrate updates in response to user needs and complex video retrieval challenges.

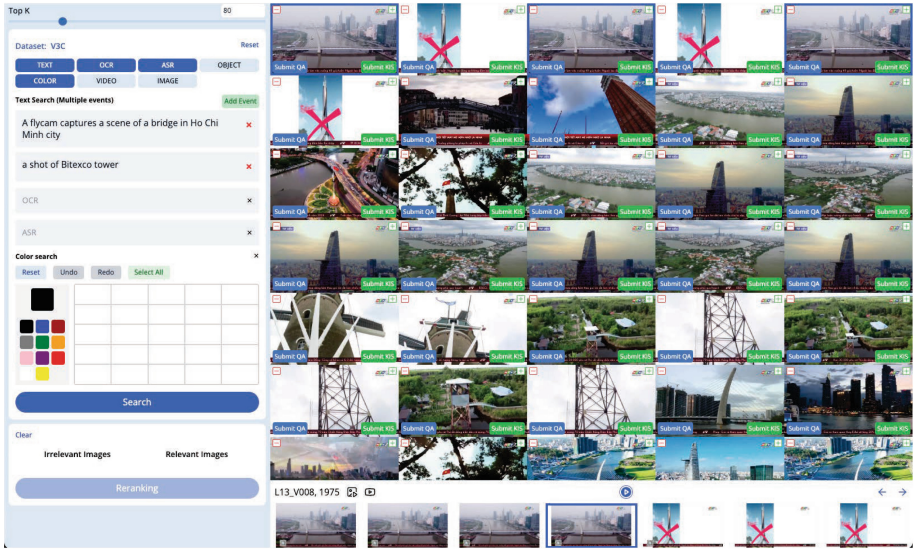


Fig. 3. The user interface of IMSEARCH 2.0

3 Conclusion

In this paper, we briefly introduce our work IMSEARCH 2.0, a video retrieval system in the competition of Video Browser Showdown 2025. Several deep VLMs

are developed from the backend with an innovative user interface, improving users' experience and interaction when performing retrieval tasks.

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